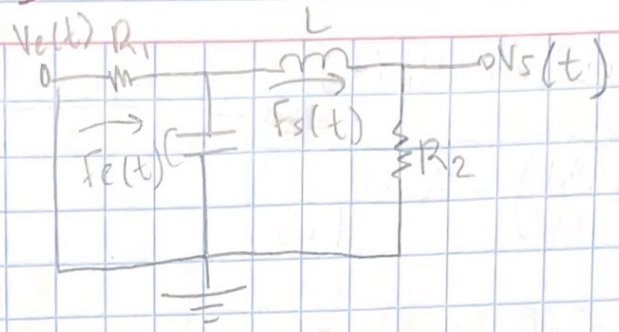
ANALISIS MATEMATICO



$$V_e(t) = R_1 F_e(t) + \frac{1}{C}\int \left[F_e(t) - F_s(t)\right] dt$$

$$V_s(t) = R_2 F_s(t)$$

$$\frac{1}{C}\int \left[F_e(t) - F_s(t)\right] dt = L\frac{dF_s(t)}{dt} + R_2 F_s(t)$$

Transformada de Laplace

$$V_e(s) = R_1 F_e(s) + \frac{F_e(s) - F_s(s)}{Cs}$$

$$\frac{F_e(s) - F_s(s)}{Cs} = Ls\, F_s(s) + R_2 F_s(s)$$

Despejamos $F_e(s)$

$$\frac{F_e(s)}{Cs} - \frac{F_s(s)}{Cs} = Ls\, F_s(s) + R_2 F_s(s)$$

$$\boxed{\frac{CLs^2 + R_2 Cs + 1}{Cs}}$$

$$\frac{F_e(s)}{Cs} = F_s(s)\left(Ls + R_2 + \frac{1}{Cs}\right)$$

$$F_e(s) = F_s(s)\,\cancel{Cs}\left(\frac{CLs^2 + R_2 Cs + 1}{\cancel{Cs}}\right)$$

$$F_e(s) = F_s(s)\left(CLs^2 + R_2 Cs + 1\right)$$

Sustituimos $F_e(t)$ en $V_e(t)$

$$V_e(s) = F_e(s)\overbrace{\left(R_1 + \frac{1}{Cs}\right)}^{\frac{R_1Cs+1}{Cs}} - F_s(s)\left(\frac{1}{Cs}\right)$$

$$V_e(s) = F_s(s)\left[\left(CLs^2 + R_2(s+1)\right)\left(\frac{R_1Cs+1}{Cs}\right) - \frac{1}{Cs}\right]$$

$$V_e(s) = F_s(s)\left[\frac{R_1C^2Ls^3 + CLs^2 + R_1R_2C^2s^2 + R_2Cs + R_1Cs + 1 - 1}{Cs}\right]$$

$$= F_s(s)\left[\frac{\cancel{Cs}(R_1CLs^2 + Ls + R_1R_2Cs + R_2 + R_1)}{\cancel{Cs}}\right]$$

$$V_e(s) = F_s(s)\left(R_1CLs^2 + s(L + R_1R_2C) + R_1 + R_2\right)$$

$$V_s(s) = \frac{\cancel{F_s(s)}\ R_2}{\cancel{F_s(s)}\ (R_1CLs^2 + s(L + R_1R_2C) + R_1 + R_2)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R_2}{R_1CLs^2 + s(L + R_1R_2C) + R_1 + R_2}$$